

Autonomous Control That Fails Safe

A reactor cascade controller that acts on its own — and trips safe in one tick.

AUTONOMOUS-MOCK

Autonomous control is the line plants won't cross

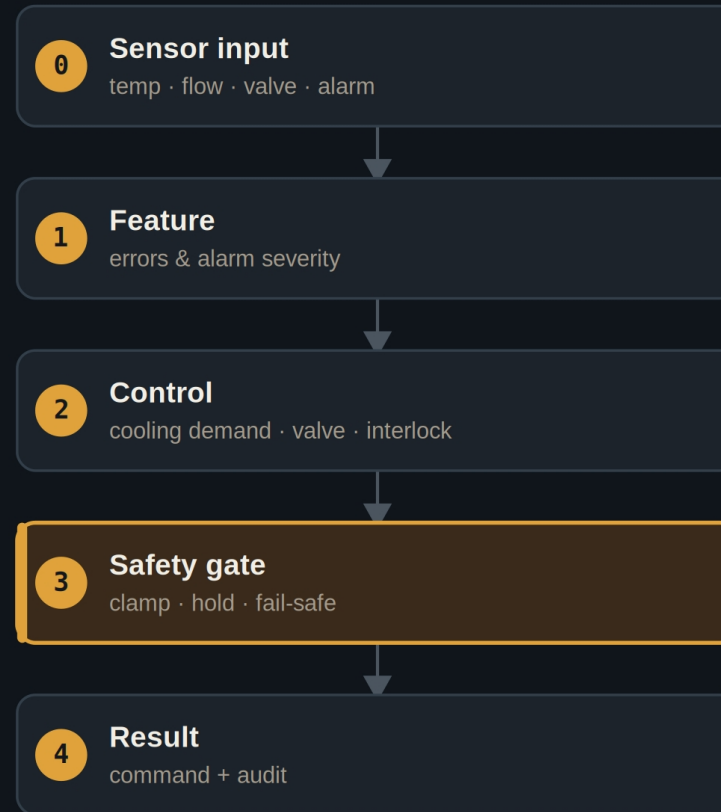
- A neural controller that can open and close valves on its own is exactly what operators fear.
- One bad command under an alarm condition is a safety incident, not a bug ticket.
- **So the loop stays manual**
and the efficiency stays on the table.

The blocker isn't the control law. It's what happens on the worst tick.

Cascade control with a fail-safe that can't be overridden

- **It acts**
a real cascade loop drives the valve toward the cooling target.
- **It yields**
a dedicated safety layer sits above the controller and can override it.
- **It records**
every command — issued, held, or vetoed — lands in an audit stream.

The safety gate sits above the controller



The safety layer can override both the control decision and the valve command.

Calm when calm, instant when not

NORMAL

- Process values in range.
- Valve command tracks the cooling target.

HIGH-TEMP ALARM

- Safety layer overrides the controller.
- Fail-safe command issued within one tick.

200 deterministic ticks

200 / 200

ticks processed, every row present
deterministic

1 tick

alarm to fail-safe at t5

Held

manual override at t10, recorded

Autonomy, with hard limits

AUTONOMOUS-MOCK Real control intent, against a simulated actuator — the full loop, no hardware risk.

Fail-safe A high-temp alarm forces a safe command, regardless of what the controller wanted.

Auditable Issued, held, and vetoed commands are all on the record.

From mock to real is one edit

- **Swap the input edge**
the mock sensor stream becomes a live OPC UA client.
- **Keep the contract**
same typed signals, same one-tick fail-safe.
- **Verify once**
the safety you proved deterministically still holds on the live process.

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A neural controller can hold the loop and still be trusted to fail safe — because the brakes are structural, not statistical.

The case for Demo 01

JNEOPALLIUM · DEMO 01

rakovpublic/jneopallium

Autonomy you can trust to stop.

Cascade control with a fail-safe wired into the decision path.

github.com/rakovpublic/jneopallium

AUTONOMOUS-MOCK · deterministic

